Lab 05 – Data Preparation

Contribution Journal

Team TheBetaThetas · ITAI 1371

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Derrick K. Gibbs-McGlaston

My main contribution to the Module 05 Data Preparation lab was completing and reviewing the data preprocessing steps in the notebook.

I worked through the required tasks and checked the results after each step. This included handling the missing values in the Age column, converting the categorical Sex and Embarked information using one-hot encoding, and standardizing the Age and Fare features.

My focus was making sure each preparation step worked correctly before moving forward. I reviewed how the dataset changed after each operation instead of only checking whether the code ran without an error.

This contribution helped make sure the data preparation portion of the notebook was completed correctly and that the resulting data was in a better form for the next stages of the machine learning workflow.

Marlon Roberts

My contribution to this lab was creating the GitHub for my team to access, I reviewed the data and completed the work needed for the module. I copied and pasted all work into one file, made sure the work was finalized, edited the document if needed, and submitted it.

Victor Djuma

For Lab 05, my contribution focused on completing the Data Preparation notebook in Google Colab and making sure each required task was completed correctly. I began by running the setup section, which loaded the Titanic dataset and displayed the missing values. This helped me identify that the dataset

contained missing values, especially in the Age column, which was one of the main problems addressed in the lab. The notebook explained that raw data often requires preprocessing before it can be used by a machine learning model.

For the first task, I worked on **handling missing values** in the Age column. I calculated the median age and used it to replace the missing Age values. After completing the imputation, I verified the result by checking the number of missing values remaining in the Age column, which became zero. This task helped me understand how imputation can preserve useful data instead of simply deleting incomplete rows.

For the second task, I completed the **One-Hot Encoding** section. I used `pd.get_dummies()` to convert the categorical Sex and Embarked columns into numerical columns. I also used `drop_first=True` as directed in the notebook. After running the code, I checked the new DataFrame and confirmed that the original categorical columns had been replaced with encoded columns such as `Sex_male`, `Embarked_Q`, and `Embarked_S`.

For the third task, I completed the **feature scaling** section. I used `StandardScaler` from scikit-learn to scale the Age and Fare columns. After running the code, I reviewed the transformed values and confirmed that Age and Fare were standardized. This helped me understand why numerical features on different scales may need to be transformed before being used by some machine learning algorithms.

I also completed the three **Knowledge Check** questions at the end of the notebook. I explained why the median is useful for imputation, how One-Hot Encoding converts categorical information into numerical form, and why Decision Tree models generally do not require feature scaling.

Finally, I reviewed the notebook to make sure the required code cells executed successfully, saved the completed Jupyter notebook, and created a PDF version of the lab. My contribution to Lab 05 included completing the required coding tasks, checking the outputs, answering the Knowledge Check questions, and preparing my completed work for submission.

Brandy Gonzales

I completed the Lab 5 Data Preparation notebook on the Titanic dataset. I filled the 177 missing Age values with the median and checked that none were left. I used one-hot encoding to turn Sex and Embarked into number columns, and I scaled Age and Fare with `StandardScaler`. I also answered all three Knowledge Check questions in my own words: median vs. mean, what one-hot encoding does

and why, and why a decision tree doesn't need scaling. Finally, I wrote my reflection on what I learned, saved as a PDF, and sent my work to our team leader Marlon Roberts, who then submits on our behalf.

Pablo Lara

For this lab, each team member completed the full notebook on their own in Google Colab, and then we combined our work and journals into one submission. My contributions were:

- **Completed the full notebook on my own.** I worked through all three coding tasks in Colab:
  - Task 1, Imputation: I calculated the median of the Age column (28.0) and used it to fill in the 177 missing ages. I then checked that no missing values remained.
  - Task 2, One-Hot Encoding: I encoded the Sex and Embarked columns using `pd.get_dummies()` with `drop_first=True`. Sex became a single `Sex_male` column, and Embarked became `Embarked_Q` and `Embarked_S`.
  - Task 3, Feature Scaling: I used `StandardScaler` to standardize the Age and Fare columns so they have a mean of 0 and a standard deviation of 1.
  - Knowledge Check: I answered the three questions on median vs. mean, one-hot encoding, and whether Decision Trees need scaling.
- **Found and fixed a problem with the notebook file.** The notebook had lost its line breaks, so every code cell was squashed onto one line and most cells gave a syntax error. I fixed each cell by putting every statement on its own line.
- **Caught a silent error and warned the team.** The final scaling cell ran without any error but did not actually scale anything, because everything after the first # was treated as a comment. I noticed something was wrong, corrected the cell, and let my team know to watch for this issue in the last code cell, including how to check their output to confirm the scaling worked.
- **Verified my results.** I re-ran the cells to make sure my outputs were correct and that the notebook worked from start to finish.